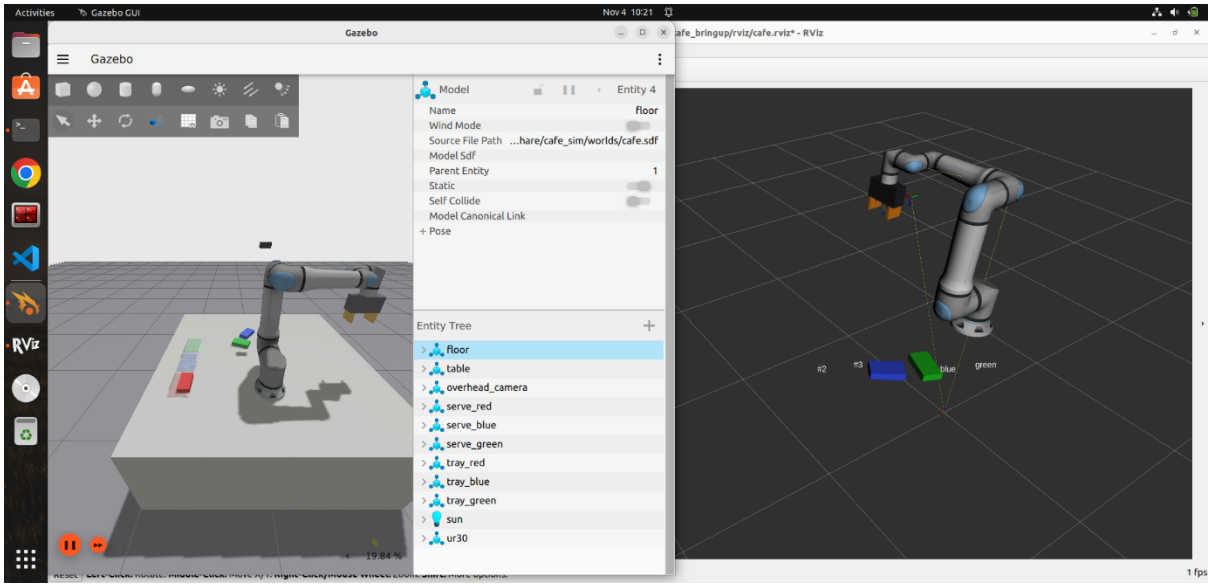
Assessment: ROS Developer (Vision-Based Manipulator System)

A camera detects coloured food trays lying randomly on a kitchen table; a UR30 with a parallel gripper

picks each tray and puts it in the serving area that belongs to its colour. Handles missing trays,

failed grasps (retry → skip), trays that are moved while running, and colour priority.



Preoject Specification

Project Developed in ROS2 humble in ubuntu 22.04

Robot – UR30

Gripper- custom 2-finger parallel gripper

Camera – RGB-D camera

Tray – Red, Blue, Green (assume square shape tray)

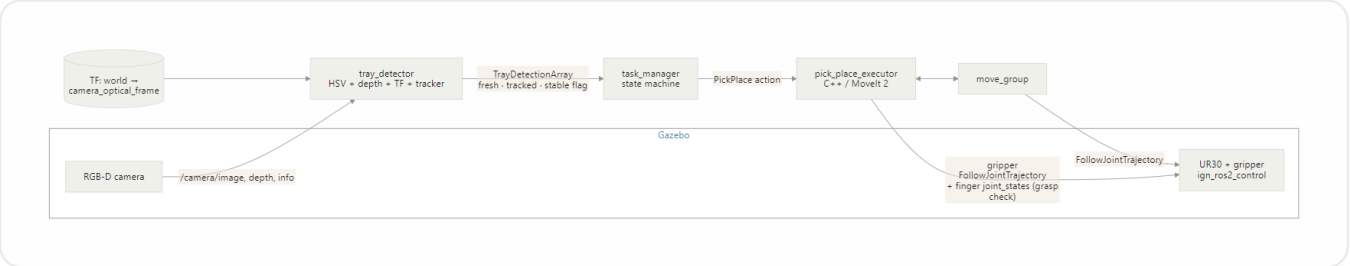
Food-package – Red Blue, Green in Square shape

Simulator - Gazebo Fortress (`ign`, default)

Planning - MoveIt 2 (OMPL RRTConnect for free motion,
Cartesian paths for approach/lift/place)

Perception - Overhead RGB-D camera,
OpenCV HSV segmentation + depth, TF2 camera→world

Architecture



Packages and Roles

Package	Role
cafe_interfaces	TrayDetection(.Array) msgs, PickPlace action (typed error codes)
cafe_sim	URDF/xacro (UR30 + gripper + ros2_control), Gazebo world, controllers, sim.launch.py, randomize_trays.py
cafe_moveit_config	RDF, KDL kinematics, OMPL, joint limits, controller mapping
tray_detector	Perception node. core.py is ROS-free and unit-tested
pick_place_executor	Pick/place action server: planning, grasp verification, recovery
task_manager	Detect → prioritise → execute → retry/skip → idle
cafe_bringup	One parameter file, **XML launch files**, RViz config

Setup

❖ Run this in terminal

```
sudo apt update

sudo apt install -y ros-humble-desktop ros-humble-moveit ros-humble-ur-description \
  ros-humble-ros-ign ros-humble-ign-ros2-control ros-humble-ros2-controllers \
  ros-humble-controller-manager ros-humble-xacro ros-humble-cv-bridge \
  ros-humble-message-filters ros-humble-tf2-ros-py python3-opencv python3-colcon-common-extensions
```

❖ UR30 must exist in your ur_description:

```
ls /opt/ros/humble/share/ur_description/config/ur30 || \
  git clone -b humble https://github.com/UniversalRobots/Universal_Robots_ROS2_Description.git ~/cafe_ur30_ws/src/ur_description_src
```

❖ **Open & extract this zip file- Souce it & then build this package**

```
cd ~/cafe_ur30_ws
source /opt/ros/humble/setup.bash
rosdep install --from-paths src --ignore-src -r -y
colcon build --symlink-install
source install/setup.bash
```

Run the project

- Gazebo + RViz + everything (~20 s to come up)

  ```
  ros2 launch cafe_bringup cafe.launch.xml
  ```

- headless Gazebo

  ```
  ros2 launch cafe_bringup cafe.launch.xml gui:=false
  ```

Checking different scenario

- new random layout

  ```
  ros2 run cafe_sim randomize_trays.py --seed 4
  ```

- move a tray (even mid-pick) -> dynamic update

  ```
  ros2 run cafe_sim randomize_trays.py --move red
  ```

- tray disappears -> idle / re-scan

  ```
  ros2 run cafe_sim randomize_trays.py --remove blue
  ```

- force failures -> retry -> skip

  ```
  ros2 param set /pick_place_executor simulate_grasp_failure true
  ros2 param set /pick_place_executor simulate_grasp_failure false
  ```

- clear skips

  ros2 service call /task_manager/reset std_srvs/srv/Trigger

- restart node to apply- nearest-first

  ros2 param set /task_manager priority_enabled false

- check

  ros2 topic echo /task_manager/stats

Note – Due to time constraint – containerized this whole application is still in process